THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS · EASY LEVEL CHAPTER TEST SET 2

Chapter 14: **Surface Area and Volume** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The total surface area of a cube of edge 7 cm is: 1

(a) 264 cm^2 **(b)** 324 cm^2 **(c)** 354 cm^2 **(d)** 294 cm^2

2. The lateral surface area of a cuboid 3 cm $\times$ 8 cm $\times$ 9 cm (height 9 cm) is: 1

(a) 198 cm^2 **(b)** 178 cm^2 **(c)** 239 cm^2 **(d)** 218 cm^2

3. The formula for the surface area of a sphere of radius r is: 1

(a) $\frac{4}{3}\pi r^3$ **(b)** πr^2 **(c)** $2\pi r^2$ **(d)** $4\pi r^2$

4. The volume of a cube is 125 cm^3. Its edge is: 1

(a) 7 cm **(b)** 6 cm **(c)** 4 cm **(d)** 5 cm

5. Take $\pi = \frac{22}{7}$. The volume of a cylinder of radius 3.5 cm and height 6 cm is: 1

(a) 207 cm^3 **(b)** 255 cm^3 **(c)** 231 cm^3 **(d)** 279 cm^3

6. Take $\pi = \frac{22}{7}$. The surface area of a sphere of radius 3.5 cm is: 1

(a) 154 cm^2 **(b)** 186 cm^2 **(c)** 170 cm^2 **(d)** 138 cm^2

7. The volume of a cuboid of length 2 cm, breadth 5 cm and height 7 cm is: 1

(a) 86 cm^3 **(b)** 78 cm^3 **(c)** 62 cm^3 **(d)** 70 cm^3

8. The volume of a cube of edge 9 cm is: 1

(a) 729 cm^3 **(b)** 656 cm^3 **(c)** 876 cm^3 **(d)** 802 cm^3

9. The total surface area of a cube of edge 8 cm is: 1

(a) 384 cm^2 **(b)** 462 cm^2 **(c)** 345 cm^2 **(d)** 423 cm^2

10. The lateral surface area of a cuboid 7 cm $\times$ 3 cm $\times$ 9 cm (height 9 cm) is: 1

(a) 161 cm^2 **(b)** 199 cm^2 **(c)** 180 cm^2 **(d)** 218 cm^2

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Find the volume and total surface area of a cube of edge 7 cm. Also find the edge of a cube whose volume is 512 cm³. 2

12. A water tank is 5 m long, 9 m wide and 2 m deep. How many litres of water does it hold when full? (1 m³ = 1000 litres) 2

13. Find the volume, total surface area and lateral surface area of a cuboid of length 9 cm, breadth 2 cm and height 8 cm. 2

14. Find the volume and total surface area of a cube of edge 9 cm. Also find the edge of a cube whose volume is 1000 cm³. 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. Find the volume, total surface area and lateral surface area of a cuboid of length 10 cm, breadth 8 cm and height 11 cm. 3

16. Take $\pi = \frac{22}{7}$. Find the curved surface area, total surface area and volume of a cylinder of radius 14 cm and height 5 cm. 3

17. Find the volume and total surface area of a cube of edge 14 cm. Also find the edge of a cube whose volume is 3375 cm³. 3

OR

Find the volume, total surface area and lateral surface area of a cuboid of length 5 cm, breadth 6 cm and height 9 cm. 3

18. Take $\pi = \frac{22}{7}$. Find the curved surface area, total surface area and volume of a cylinder of radius 14 cm and height 11 cm. 3

OR

Find the volume and total surface area of a cube of edge 18 cm. Also find the edge of a cube whose volume is 6859 cm³. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. Take $\pi = \frac{22}{7}$. Find the curved surface area, total surface area and volume of a cylinder of radius 14 cm and height 12 cm. 5

OR

How many cubes of edge 2 cm can be cut from a cuboid measuring 4 cm × 6 cm × 8 cm? Find the leftover volume if each cube is cut and the cuboid is fully used. 5

20. Take $\pi = \frac{22}{7}$. Find the curved surface area, total surface area and volume of a cylinder of radius 10.5 cm and height 8 cm. 5

OR

How many cubes of edge 4 cm can be cut from a cuboid measuring 20 cm × 24 cm × 16 cm? Find the leftover volume if each cube is cut and the cuboid is fully used. 5

END OF QUESTION PAPER Total: 40 Marks

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